PHASE 3 – PROJECT DESIGN

Automated Standard Laptop Procurement System

The Project Design phase describes the structure and working model of the proposed automated standard laptop procurement system. It explains how employees submit requests, how requests are validated and processed, how IT/Procurement personnel handle the requests, and how the final status is communicated to the employee. The design provides a clear blueprint for the development and implementation stages.

1. System Design

The system is designed to automate the complete standard laptop ordering workflow. The proposed system provides a structured process through which an employee can submit a laptop request and track its progress. The request is validated before it is processed, and the IT/Procurement team handles the necessary verification and order-processing activities.

The system design focuses on the following objectives:

• Provide a simple method for employees to submit standard laptop requests.

• Collect the required employee and laptop information in a structured format.

• Validate request details before further processing.

• Check whether the requested laptop belongs to the standard laptop category.

• Route valid requests to the appropriate IT/Procurement team.

• Allow authorized users to process orders and update request status.

• Provide request tracking so employees can view the progress of their requests.

• Maintain organized procurement information for monitoring and administration.

• Reduce repetitive manual work and improve consistency in the procurement workflow.

2. Basic Workflow

The basic workflow represents the sequence followed by a laptop request from submission to completion. Each stage has a specific purpose and the request moves to the next stage after the required information or action is completed.

Employee

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Laptop Request

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Request Validation

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Check Standard Laptop

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Process Request

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IT / Procurement Team

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Order Processing

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Request Status

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Employee

2.1 Workflow Description

Stage	Description
1. Employee	The employee starts the process by identifying the need for a standard laptop and submitting a request through the system.
2. Laptop Request	The employee enters required details such as employee information, department, laptop requirement, model, business justification, and request date.
3. Request Validation	The submitted information is checked for completeness and correctness. Missing or invalid information can be identified before processing.
4. Check Standard Laptop	The system verifies whether the requested laptop model or requirement matches the organization's standard laptop options.
5. Process Request	A valid request is made available for processing by the responsible IT/Procurement team.
6. IT / Procurement Team	Authorized personnel review the request, verify the information, and take the required procurement action.
7. Order Processing	The procurement team processes the laptop order and records the relevant order information.
8. Request Status	The status of the request is updated according to its current stage, such as submitted, under review, processing, or completed.
9. Employee	The employee can view the latest request status and receive the final order information through the system.

3. Main Modules

The system is divided into major modules based on the type of user and the activities performed by that user. This modular design makes the system easier to understand, develop, test, and maintain.

Module	Purpose	Main Functions
Employee Module	Allows employees to create and monitor their laptop requests.	• Submit laptop request • View submitted request • Track request status • View order information • Provide required request details
IT / Procurement Module	Handles verification, processing, and updating of laptop requests.	• View requests • Verify request information • Check standard laptop requirement • Process orders • Update request status • Record order details
Administrator Module	Provides administrative control and monitoring of the procurement system.	• Manage users • Manage requests • Monitor procurement activities • Review system records • Support access and process management

3.1 Employee Module

The Employee Module is the entry point for employees who need a standard laptop. The employee provides the required information and submits the request. After submission, the employee can view the request details and monitor the current status without repeatedly contacting the procurement team.

Main activities:

• Submit laptop request

• View request details

• Track request status

• View order information

• Provide business justification

3.2 IT / Procurement Module

The IT/Procurement Module is responsible for handling submitted requests. Authorized team members can view incoming requests, verify the submitted information, check the standard laptop requirement, process the order, and update the status. This module forms the main processing part of the procurement workflow.

• View submitted requests

• Verify employee and request details

• Check standard laptop model

• Process laptop orders

• Update request status

• Maintain order information

3.3 Administrator Module

The Administrator Module provides higher-level control over the system. Administrators can manage users and requests and monitor procurement activities. This module supports organized management of the overall system.

• Manage users

• Manage requests

• Monitor procurement activities

• Review system information

• Support administrative control

4. Input Design

Input design defines the information collected from users when a laptop request is submitted. The fields should be clear, relevant, and sufficient for request validation and procurement processing.

Input Field	Purpose
Employee Name	Identifies the employee submitting the request.
Employee ID	Provides a unique reference for the employee.
Department	Identifies the employee's department or working unit.
Laptop Requirement	Describes why a laptop is required and the intended use.
Standard Laptop Model	Specifies the standard laptop model requested.
Business Justification	Provides the reason or business need for the request.
Request Date	Records the date on which the request was submitted.

4.1 Input Validation

Before a request moves to processing, the system should check whether the required information has been entered correctly. Validation can include checking mandatory fields, employee identification, valid request information, acceptable laptop models, and the presence of a business justification. Validation helps reduce incorrect or incomplete requests.

5. Output Design

Output design defines the information presented to the employee, IT/Procurement team, and administrator after a request is submitted and processed. The outputs help users understand the current state of the request and the result of procurement activities.

Output	Description
Request Number	A unique reference number used to identify and track the laptop request.
Request Status	Shows the current stage of the request, such as submitted, under review, processing, or completed.
Laptop Order Information	Displays relevant information about the laptop order after procurement processing.
Approval / Processing Information	Provides information about the request's processing or approval stage.
Final Order Status	Shows the final result of the procurement request after processing is completed.

6. Functional Design

The functional design explains what each major part of the system is expected to do. The system begins with employee input, performs validation, routes the request for processing, and returns status information to the employee.

Function	Expected Behaviour
Request Submission	Employee enters the required information and submits the laptop request.
Validation	System checks whether required information is complete and valid.
Standard Laptop Check	System verifies that the selected laptop belongs to the defined standard category.
Request Processing	Valid requests are made available to the IT/Procurement team.
Order Processing	IT/Procurement team performs the necessary order-processing activities.
Status Update	Authorized users update the request according to its current processing stage.
Request Tracking	Employee can view the current request status using the request reference.
Administration	Administrator manages users, requests, and monitoring activities.

7. Non-Functional Design Considerations

Apart from functional requirements, the system design should also consider qualities that improve the overall usefulness and reliability of the application.

• **Usability:** Forms and screens should be simple and easy to understand.

• **Reliability:** The workflow should process valid requests consistently.

• **Security:** Only authorized users should access administrative and procurement functions.

• **Accuracy:** Input validation should reduce incorrect request information.

• **Maintainability:** The modular design should make future changes easier.

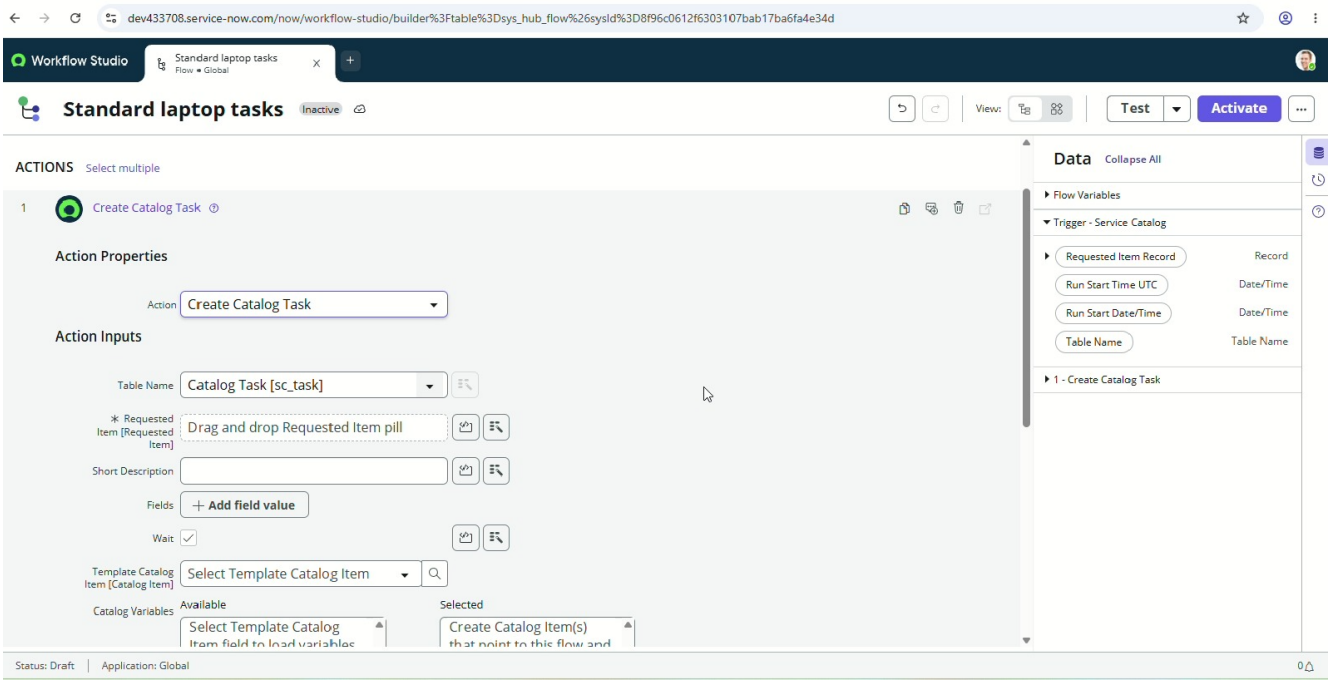
• **Traceability:** Requests should have identifiable numbers and status information for tracking.

8. Design Summary

The proposed design provides a structured and automated flow for standard laptop procurement. Employees submit requests through the Employee Module, the system validates and checks the request, and the IT/Procurement Module processes valid requests. The Administrator Module provides management and monitoring capabilities. Input and output designs ensure that important request information is collected and useful processing results are provided. This design serves as the foundation for the development and testing phases of the project.

9. Conclusion

The Project Design phase converts the project requirements into a clear system structure and workflow. By defining the modules, process flow, inputs, outputs, functional activities, and design considerations, the team obtains a clear blueprint for implementation. The design also supports easier testing and future maintenance of the automated standard laptop procurement system.


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Workflow Studio
Standard laptop tasks
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Standard laptop tasks
Inactive
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Activate
ACTIONS
Select multiple
1
Create Catalog Task
Action Properties
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Table Name
Catalog Task [sc_task]
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Collapse All
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Table Name
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1 - Create Catalog Task
Status: Draft
Application: Global